

Deep Spectrum Sensing Under Realistic RF Conditions: A Cross-Dataset Study

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Abstract—Spectrum sensing constitutes a core function in cognitive radio systems. Traditional sensing methods often suffer from limited robustness due to noise uncertainty and their dependence on prior signal knowledge. These limitations have motivated the exploration of data-driven techniques capable of learning discriminative features directly from raw measurements. This work investigates deep learning-based spectrum sensing within a unified binary classification framework. Four deep learning architectures, including a convolutional neural network (CNN), a long short-term memory (LSTM) network, a gated recurrent unit (GRU) network, and a lightweight phase-encoded two-stream GRU, are evaluated using both raw in-phase/quadrature (I/Q) samples and time-frequency representations. Performance is assessed under low signal-to-noise ratio (SNR) conditions and realistic channel impairments using metrics such as probability of detection, accuracy, and F1-score. The findings demonstrate the potential of lightweight deep learning approaches for practical spectrum sensing in next-generation cognitive radio and real-time RF monitoring systems.

Index Terms—spectrum sensing, cognitive radio, deep learning, convolutional neural networks, recurrent neural networks.

I. INTRODUCTION

The rapid growth of wireless connectivity driven by 5G deployment and the emergence of 6G services has intensified pressure on scarce radio-frequency (RF) spectrum resources [1], [2]. Cognitive Radio (CR) improves spectrum utilization by enabling secondary users (SUs) to opportunistically access underutilized frequency bands while protecting primary users (PUs) [2], [3]. Reliable spectrum sensing is therefore essential, particularly under low signal-to-noise ratio (SNR) and practical RF impairments [1], [4], [5].

Conventional sensing techniques include energy detection, matched filtering, and cyclostationary feature detection [4], [5]. Learning-based methods have been explored to improve robustness. Classical machine learning (ML) relies on hand-crafted features and standard classifiers [1], [6], but generalization is often constrained by feature design and channel variability [1], [5]. Deep learning (DL) enables end-to-end sensing from raw in-phase/quadrature (I/Q) samples or time-frequency features [7]–[9]. CNNs capture spectral structures, whereas recurrent models (LSTM/GRU) exploit temporal dependencies and phase dynamics [8], [10]. Hybrid and physical-layer DL frameworks further improve robustness under fading and carrier frequency offset (CFO) impairments [7], [11].

Despite recent progress, many DL-based sensing models remain computationally demanding and are typically evaluated using a single dataset, limiting cross-dataset generalization and feasibility for edge deployment [1], [8]. Meanwhile, many public RF datasets primarily target modulation recognition rather than a unified binary sensing task [11], [12], complicating benchmarking and reducing comparability across studies.

To address these limitations, this paper investigates lightweight DL architectures for binary spectrum sensing across four heterogeneous RF datasets: RML2016b [12], RML2018a [11], HisarMod2019 [13], and PanoRadioHF [14]. Modulated samples are treated as occupied channels, while idle-channel samples are detected as noise-only observations, enabling consistent cross-dataset evaluation. The main contributions are summarized as follows:

- A unified binary spectrum sensing formulation applied across four heterogeneous RF datasets for consistent cross-dataset evaluation under low-SNR conditions and diverse channel impairments.
- Comparative evaluation of four DL architectures (CNN, LSTM, GRU, and PETCGDNN) using raw I/Q samples and time-frequency representations.
- A combined performance and computational complexity analysis using probability of detection, F1-score, accuracy, NMSE, AUC, parameter count, number of operations, and CPU-only inference latency.
- An experimental analysis demonstrating that recurrent architectures, which exploit temporal correlations, tend to deliver more stable detection performance across the evaluated datasets under challenging channel conditions.

The source code of this work is available on: <https://github.com/INRS-ECCoLe/SpectrumSensingNoisy>.

II. RELATED WORK

A. Classical Spectrum Sensing

Classical spectrum sensing has been widely investigated for CR systems, including energy detection, matched filtering, and cyclostationary feature detection [4], [5]. Energy detection methods are simple to implement and do not require PU waveform knowledge; however, they are typically sensitive to noise uncertainty and suffer from SNR walls in real-world receivers

[5], [15]. Matched filtering achieves optimal detection given perfect prior knowledge of the PU signal, which is generally unavailable in non-cooperative environments [4]. In contrast, cyclostationary methods provide robustness against noise and modulation variability, but their computational complexity limits real-time deployment [5], [6].

B. Machine Learning-Based Sensing

To improve robustness, ML-based sensing has leveraged handcrafted feature pipelines (energy statistics, spectral descriptors, cyclostationary features) coupled with conventional classifiers [1], [6]. While ML approaches often outperform purely analytical detectors in moderate SNR regimes, their generalization remains constrained by feature design choices and distribution shifts across channel conditions [1], [5].

C. Deep Learning-Based Sensing

DL-based sensing enables end-to-end learning directly from raw I/Q data or time–frequency representations, removing the need for manual feature engineering [7]–[9]. CNNs typically operate on spectrogram or 2D representations derived from I/Q samples over a temporal window, whereas LSTM/GRU-based models capture temporal dependencies and phase-related dynamics that are critical under fading and CFO [8], [10]. More recent physical-layer DL frameworks and hybrid designs further improve robustness to realistic impairments [10], [11]. However, many existing DL solutions remain computationally heavy and are typically evaluated on a single dataset, limiting conclusions about cross-dataset generalization and deployability on edge devices [1], [8]. These limitations motivate this work, which focuses on lightweight DL architectures and cross-dataset evaluation under low-SNR conditions.

III. RF DATASETS AND PROBLEM SETUP

Four RF datasets are used for evaluations in this work. Although originally developed for modulation recognition, these datasets are reformulated for binary spectrum sensing to enable cross-dataset evaluation under diverse channel conditions.

A. RF Datasets

RML2016b: contains digital and analog modulated signals generated under additive white Gaussian noise (AWGN) over a wide SNR range. Each sample consists of 128×2 I/Q samples, representing a short observation window suitable for lightweight spectrum sensing evaluation.

RML2018a: includes realistic channel impairments such as multipath fading, CFO, and hardware distortions. Signals are represented by 1024×2 I/Q samples, providing longer observations and increased channel complexity compared to RML2016b.

HisarMod2019: is generated using a realistic communication system simulator and models multipath propagation, Doppler effects, and hardware non-idealities. The dataset consists of 1024×2 I/Q samples and reflects time-varying channel conditions, making it well suited for evaluating temporal deep learning models.

PanoRadioHF: consists of real-world high-frequency (HF) radio signals using 2048×2 I/Q samples. HF channels are inherently characterized by strong frequency selectivity, ionospheric propagation effects, and non-stationary noise, which make this dataset significantly more challenging and less idealized than commonly used synthetic benchmarks.

B. Reformulation for Spectrum Sensing

A unified binary formulation is adopted across all datasets, where all modulated signals are labeled as occupied channels ($\mathcal{H}_1$). To ensure consistent cross-dataset evaluation and prevent the models from exploiting trivial dataset-specific noise floor differences, idle-channel samples ($\mathcal{H}_0$) are not represented by AWGN alone. Instead, they are synthetically generated using a standardized complex RF background simulator, as described in Section IV-A. In addition, all signals are power-normalized to maintain consistency across datasets and SNR levels.

IV. METHODOLOGY

This section presents the proposed deep learning-based spectrum sensing framework, including signal preprocessing, feature representations, model architectures, and the training and evaluation protocol. Fig. 1 provides an overview of the development and evaluation phases considered in this work.

A. Problem Formulation

Spectrum sensing is formulated as a binary hypothesis testing problem, where the received baseband signal is classified as either idle ($\mathcal{H}_0$) or occupied ($\mathcal{H}_1$). In practical RF environments, however, idle channels rarely correspond to ideal thermal noise or pure AWGN. Using pure AWGN for the $\mathcal{H}_0$ class can lead deep learning models to achieve artificially high accuracy by exploiting simple cues such as spectral flatness or signal energy, thereby masking their true generalization capability under realistic impairments.

To address this limitation and ensure rigorous cross-dataset evaluation, a Complex RF Background Simulator is uniformly applied across all datasets. The synthesized $\mathcal{H}_0$ (signal absent) class consists of a dynamic mixture of: (i) AWGN defining the baseline noise floor, (ii) colored noise generated via low-pass filtering to emulate RF front-end imperfections, (iii) narrowband interference (NBI) modeled as random continuous wave (CW) tones to represent adjacent channel leakage and spurious emissions, and (iv) impulsive noise consisting of high-amplitude, sparse transient spikes representing environmental static and switching noise.

This more realistic complex idle-signal simulator was developed in Python. NumPy was used to generate the baseline AWGN, random impulsive spikes, and continuous-wave tones for NBI, while SciPy signal processing tools were used to apply low-pass filtering and generate the colored-noise component.

Modulated samples from the considered datasets are labeled as $\mathcal{H}_1$ (signal present). By injecting structured, non-Gaussian noise and interference into the $\mathcal{H}_0$ baseline, we substantially

reduce the possibility of trivial energy-based decision shortcuts. Consequently, the models are forced to extract more informative temporal, spectral, and phase-related features inherent to the modulated signals. Because this complex baseline is standardized across all four datasets, the resulting cross-dataset performance metrics accurately reflect the models' true generalization capabilities against channel impairments rather than variations in dataset SNR definitions or simplified noise floors.

B. Signal Preprocessing

All received signals are represented as complex baseband I/Q samples. To ensure fair comparison across datasets and SNR levels, each sample is power-normalized prior to feature extraction. This normalization prevents bias toward high-SNR signals and stabilizes neural network training.

C. Input Feature Representations

Two complementary feature representations are considered. First, raw I/Q sequences of fixed length are directly provided to recurrent and complex-valued neural networks to preserve temporal and phase information. Second, short-time windows of I/Q samples are reshaped into 2D inputs for convolutional neural networks, enabling the extraction of local spatio-temporal patterns.

D. Deep Learning Architectures

Four neural architectures are evaluated in this study, each previously introduced and optimized in the literature for RF signal recognition tasks. A convolutional neural network (CNN), derived from the work of Tekbıyık et al. [16], is employed to learn local spectral features from time–frequency representations. Recurrent architectures, including long short-term memory (LSTM) and gated recurrent unit (GRU) networks, adapted from the works of Rajendran et al. [17] and Hong et al. [18], respectively, are utilized to model temporal dependencies in sequential in-phase and quadrature (I/Q) samples. In addition, a lightweight phase-encoded two-stream complex-valued GRU deep neural network (PETCGDNN) [19] is incorporated to explicitly preserve phase information while reducing computational complexity. All evaluated models output the posterior probability of channel occupancy. A summary of the corresponding network architectures is provided in Table I. All models were implemented, trained, and evaluated using the PyTorch framework.

The selection of these architectures is motivated by the strong heterogeneity of the considered datasets (RML2016b, RML2018a, HisarMod2019, and PanoRadioHF), which differ in input sequence length (128×2 , 1024×2 , and 2048×2), SNR definitions, sampling rates, and propagation conditions ranging from synthetic AWGN to real-world HF ionospheric channels. Rather than a limitation, this diversity serves as a rigorous stress test for architectural generalization. By evaluating all models across this broad parameter space without dataset-specific hyperparameter tuning, we assess their structural robustness. In particular, the recurrent (LSTM, GRU, and

TABLE I
SUMMARY OF MODEL ARCHITECTURES

Model	Input Representation	Core Layers & Dimensions
CNN	I/Q samples reshaped as 2D input	4 × Conv2D layers: (256 → 128 → 64 → 64)
		Kernel size: (2 × 8) 4 × Max-Pooling layers (1 × 2) 2 × FC layers (8192 → 128 → Classes)
LSTM	Sequential I/Q samples	2 × LSTM layers Hidden units: 128 per layer FC layer (128 → Classes)
GRU	Sequential I/Q samples	2 × GRU layers Hidden units: 128 per layer FC layer (128 → Classes)
PETCGDNN	I/Q samples with explicit phase handling	Phase estimation linear layer (2048 → 1)
		2 × Conv2D layers (75 → 25 filters) Kernel sizes: (8 × 2), (5 × 1) 1 × GRU layer (128 units) FC layer (128 → Classes)

PETCGDNN) architectures examine the impact of temporal correlation modeling on detection performance across diverse scenarios, while the CNN architecture evaluates the effectiveness of spatial and short-time feature extraction.

E. Training and Evaluation Protocol

Models are trained using supervised learning with a binary cross-entropy loss function. Training and testing are conducted across multiple SNR levels to evaluate robustness under low-SNR conditions. Performance is assessed under various channel impairments, including fading and carrier frequency offset. Evaluation metrics include probability of detection, accuracy, F1-score, and area under the receiver operating characteristic curve. This unified evaluation protocol enables consistent comparison across architectures and datasets.

F. Cross-Dataset Evaluation Protocol

To substantiate our cross-dataset generalization claims, a strict and uniform evaluation protocol was consistently applied across all experiments. First, the identical Complex RF Background Simulator was used to generate the idle-channel ($\mathcal{H}_0$) class for all four datasets, ensuring a consistent interference baseline. Second, all signals underwent uniform power normalization to prevent models from exploiting dataset-specific amplitude scaling or inherent dataset power disparities. This uniform evaluation setup ensures that the observed performance variations across datasets are not attributable to inconsistencies in the evaluation procedure.

V. RESULTS AND DISCUSSION

In this section, we evaluate the performance of the selected deep learning architectures for spectrum sensing.

A. Performance Evaluation Across Datasets

The spectrum sensing performance is evaluated across four highly heterogeneous datasets, with results presented in Fig. 2 to Fig. 5 for different datasets. The primary metrics include Probability of Detection (P_d or Recall), Probability of False Alarm (P_{fa}), F1-Score, and overall Accuracy.

RML2016b: Analyzing the results in Fig. 2 for RML2016b tests, the models demonstrate strong resilience against the complex interference baseline. Even under severe noise conditions (below -10 dB SNR), baseline architectures like the

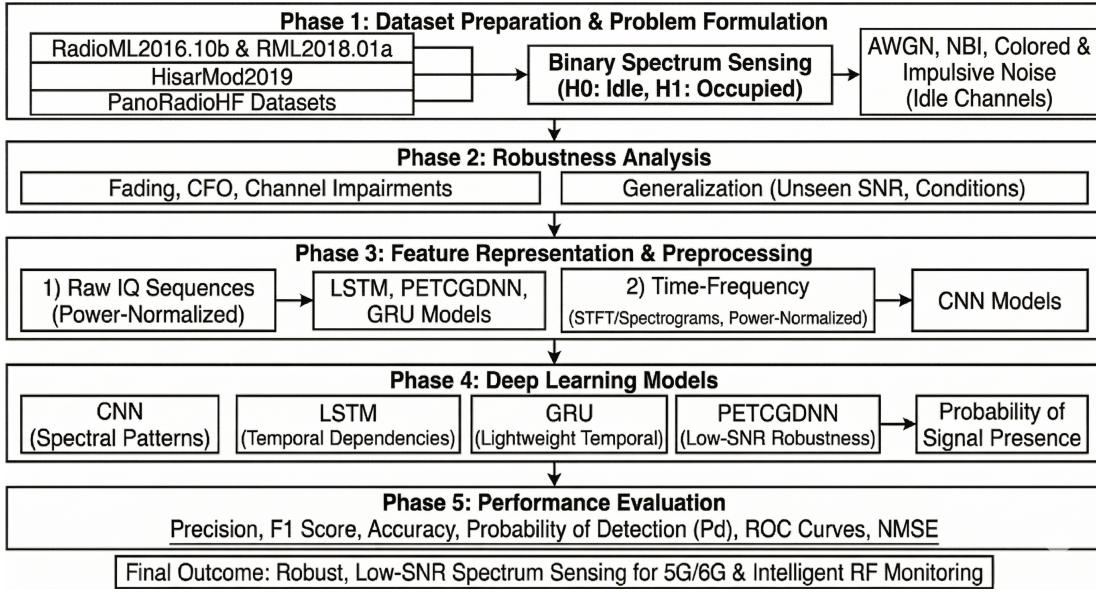


Fig. 1. Six-phase DL-based spectrum sensing evaluations, illustrating the complete processing pipeline from dataset preparation to detection decision.

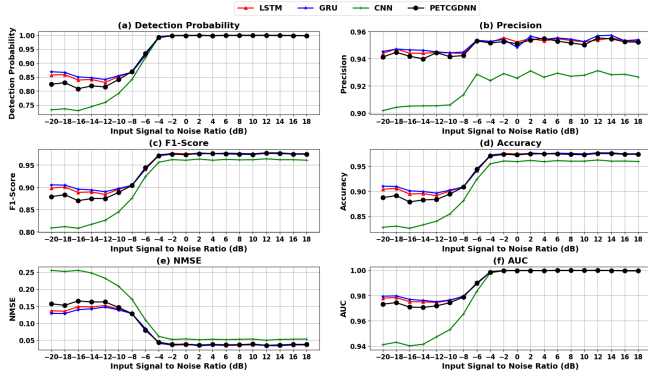


Fig. 2. Performance on RML2016b dataset (128×2 I/Q samples, AWGN channel): (a) Probability of detection (P_d), (b) Precision, (c) F1-Score, (d) Accuracy, (e) NMSE, and (f) AUC across SNR range from -20 to $+18$ dB.

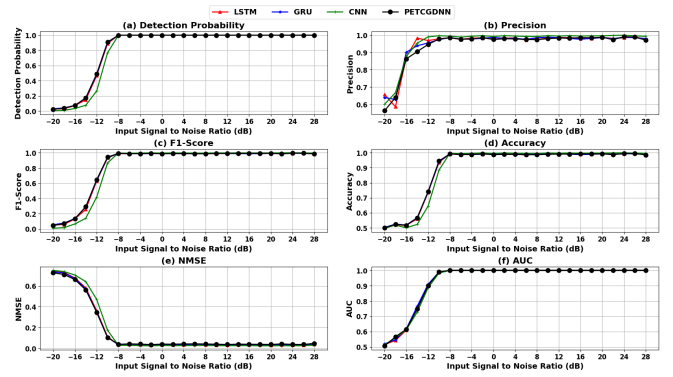


Fig. 3. Performance metrics on RML2018a dataset (1024×2 I/Q samples) under realistic channel impairments, including multipath fading, CFO, and timing offsets evaluated from -20 to 28 dB SNR.

LSTM achieve an accuracy of approximately 90% with a P_d of over 80%. The PETCGDNN and CNN architectures maintain superior precision and P_d across the SNR spectrum, effectively isolating the short-burst modulation features from the synthesized NBI and impulsive spikes. The Area Under the Curve (AUC) remains exceptionally high (>0.97) even in the negative SNR regime.

RML2018a: Results in Fig. 3 illustrate significant detection challenges at low SNRs, primarily because the longer observation windows dilute localized signal energy relative to the continuous-wave interference present in the $\mathcal{H}_0$ class. At -20 dB, standard recurrent models like the LSTM exhibit an accuracy near 50% with a P_d of just 2.5%, indicating a complete breakdown where the model struggles to differentiate the signal from the background noise. However, as the SNR approaches 0 dB and enters the positive regime, the results rapidly improve. The PETCGDNN recovers remarkably,

achieving a P_d of 1.0 (100%) and an accuracy of $> 98\%$ at $+6$ dB, demonstrating its capacity to leverage phase-encoded geometric features to lock onto the signal despite heavy background coloration.

HisarMod2019: As shown in Fig. 4, all models achieve exceptionally high performance across the entire evaluated SNR range. For example, the LSTM attains 99.3% accuracy and a P_d of 98.9% even at -20 dB, while PETCGDNN maintains near-perfect F1-scores (> 0.998) at all tested SNR levels. These excellent results suggest that the inherent statistical properties, pulse shaping, and modulation constellations present in the HisarMod samples are highly distinct from our synthesized NBI and colored noise, making them easily separable for deep learning models without relying on raw power disparities.

PanoRadioHF: As shown in Fig. 5, the models achieve excellent detection performance on PanoRadioHF, which pro-

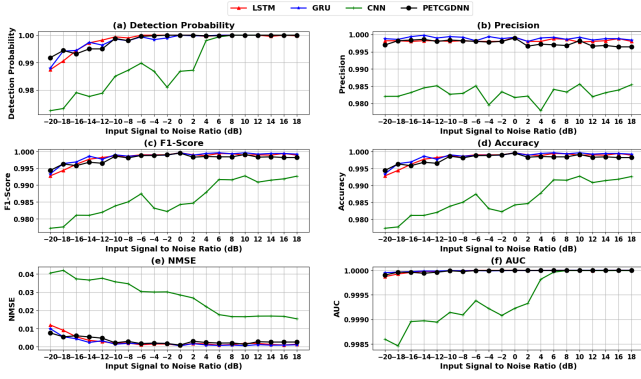


Fig. 4. Performance evaluation on HisarMod2019 dataset (1024×2 I/Q samples) with realistic multipath, Doppler effects, and hardware non-idealities from -20 to 18 dB SNR.

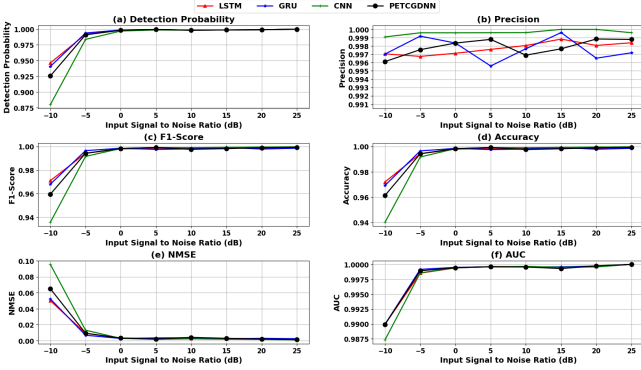


Fig. 5. Spectrum sensing results on PanoRadioHF dataset (2048×2 I/Q samples) demonstrating robustness to HF propagation effects, ionospheric distortions, and non-stationary noise from -10 to $+25$ dB SNR.

vides a critical evaluation setting involving real-world over-the-air HF ionospheric propagation. At 0 dB, all four models reach $> 99\%$ accuracy. The successful classification of real HF signals against a complex synthetic baseline indicates that all models are able to learn domain-invariant phase and frequency transition patterns.

B. Analysis

Across all four datasets, several consistent trends can be observed when evaluating the architectures against the complex RF background.

First, recurrent and phase-aware architectures (LSTM, GRU, and PETCGDNN) generally outperform CNN-based models in low-SNR regimes on the RML2016b and PanoRadioHF datasets, highlighting the importance of temporal modeling for robust spectrum sensing under challenging noise conditions and impulsive transients. In contrast, CNN-based approaches achieve competitive performance on the RML2018a and HisarMod2019 datasets, indicating that spatial-spectral representations can remain effective for spectrum sensing even in time-varying and frequency-selective channel environments.

Second, temporal feature extraction through recurrent layers contributes to improved robustness across datasets. The stable

TABLE II
MODEL COMPLEXITY AND INFERENCE LATENCY FOR DIFFERENT DATASETS

Dataset	Input Size	Model	Parameters	FLOPs	Latency (ms)
RML2016b	128×2	CNN	0.86M	157.6M	0.26
		LSTM	0.20M	50.7M	0.22
		GRU	0.15M	38.0M	1.23
		PETCGDNN	0.07M	16.3M	0.56
RML2018a & HisarMod2019	1024×2	CNN	1.77M	1.26B	1.64
		LSTM	0.20M	405.8M	1.08
		GRU	0.15M	304.3M	7.65
		PETCGDNN	0.07M	141.0M	3.98
PanoRadioHF	2048×2	CNN	2.82M	2.5B	3.32
		LSTM	0.20M	811.6M	2.88
		GRU	0.15M	608.7M	18.50
		PETCGDNN	0.08M	283.4M	8.79

performance of GRU, LSTM, and PETCGDNN across heterogeneous propagation environments (from the synthesized complex RF background to real-world ionospheric HF channels) confirms that exploiting temporal correlations typically leads to improved performance in practical spectrum sensing.

Third, when computational and memory complexity are taken into account, as summarized in Table II, LSTM and GRU offer a favorable performance–complexity trade-off. The PETCGDNN achieves comparable accuracy, but with significantly lower memory demands, making it well-suited for deployment in memory-constrained applications.

C. Computational Complexity and Inference Analysis

Beyond detection performance, computational and memory complexity are key considerations in spectrum sensing due to its stringent latency requirements. Table II summarizes the complexity of the evaluated models in terms of parameter count, floating-point operations (FLOPs), and inference latency for three input sizes (128×2 , 1024×2 , and 2048×2), corresponding to the three datasets considered in this study. Inference latency was measured using CPU-only execution on a workstation equipped with an Intel Core i7-13700 processor and 32 GB of DDR5 memory.

CNN-based models achieve low inference latency due to highly parallel convolutional operations. However, this comes at the cost of significantly higher parameter counts and computational complexity, particularly as input size increases from 128×2 to 2048×2 samples.

In contrast, recurrent architectures maintain a constant number of parameters, while their computational cost and inference latency scale with the input sequence length. Although the results in Section V-A indicate comparable detection performance for GRU- and LSTM-based models, the GRU architecture incurs the highest inference latency, reaching 18.5 ms for 2048×2 inputs under CPU-only execution. While this observation may appear counterintuitive, given that GRU architectures are commonly regarded as computationally lighter than LSTMs and have lower parameter and operation counts (as summarized in Table II), it can be attributed to more efficient gate fusion and backend-level optimizations in the PyTorch LSTM implementation. This observation suggests that explicitly modeling longer-term temporal dependencies, as

enabled by LSTM, does not yield a substantial performance advantage for the considered spectrum sensing task.

Finally, PETCGDNN achieves a significant reduction in model size, with only 0.07–0.08 M parameters, and requires fewer operations (16.3–283.4 M) compared to CNN-based architectures. However, it exhibits moderately higher inference latency (0.56–8.79 ms), likely due to the reduced parallelism of its recurrent components relative to convolutional layers.

D. Discussion

The results of this study validate the necessity of rigorous model design and learning methodologies in deep spectrum sensing. By using a Complex RF Background Simulator that generates realistic interference and noise components, we effectively mitigated trivial energy-based decision shortcuts.

The parametric diversity of the four chosen datasets—varying from 128 to 2048-sample inputs, different sampling rates, and disparate propagation models—served as an intentional stress test for model generalization. The fact that models such as the CNN maintained acceptable AUC and $F1$ -score performance across this heterogeneous setting demonstrates meaningful generalization capability, while LSTM, GRU, and PETCGDNN generally emerged as the strongest performers, likely due to their ability to capture temporal correlations. The varying performance obtained in different datasets reflects the statistical complexities of the underlying signal modulations rather than an artifact of simplistic noise modeling.

Beyond detection accuracy, practical deployment in cognitive radio systems requires careful consideration of computational complexity and inference latency. Although LSTM- and GRU-based models provide strong detection performance, their sequential processing leads to increased latency and computational cost, particularly for longer input sequences such as the 1024- and 2048-sample records in RML2018a and PanoRadioHF. In contrast, the CNN benefits from highly parallelizable operations, enabling faster inference suitable for rapid spectrum scanning, albeit at the cost of higher memory usage. PETCGDNN offers a compelling compromise, combining a lightweight phase-aware architecture with improved efficiency and robust performance under complex RF backgrounds.

VI. CONCLUSION

This paper investigated deep learning-based spectrum sensing using a unified binary formulation across four heterogeneous RF datasets. A comparative evaluation of four deep learning architectures shows that phase-aware and recurrent models achieve improved detection performance in low-SNR regimes for datasets affected by AWGN and multipath fading. These results indicate that modeling temporal correlations contributes to enhanced robustness under fading conditions and carrier frequency offset impairments. The CNN-based model provides competitive performance at moderate to high SNRs while offering lower inference latency and better suitability for efficient hardware implementation. Future work will focus on real-time over-the-air SDR validation, extension to

wideband multi-band spectrum sensing, and exploration of Transformer-based architectures for long-sequence modeling. We also plan to investigate model compression to support efficient deployment on edge and IoT devices.

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